

CHAPTER 12

Chi-square tests

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- 12.1** understand the chi-square goodness-of-fit test, use it to test whether a given distribution is similar to a specified distribution (such as the Poisson distribution or normal distribution) and use it to test a hypothesis about a population proportion
 - 12.2** understand the chi-square test of independence and use the chi-square test of independence to determine whether two variables are independent.
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Introduction

This chapter explains hypothesis tests for categorical data based on one or two samples. We consider two techniques. The first is the **chi-square goodness-of-fit test**. This helps us to determine whether a sample could have come from a given type of population distribution. This is a useful test because we often encounter real-life data which we need to check to determine what distribution it follows, such as normal, binomial or Poisson. The tests we have done so far assume that the variables are normally distributed. Nonparametric statistical techniques do not require the assumption of a normal distribution. Knowing the distribution of a given sample can help us to decide what type of hypothesis test to use.

The second technique we look at in this chapter is the **chi-square test of independence**, which tests whether there is a relationship (or dependence) between two categorical variables. There are instances in business when a manager may need to make a decision based on categorical data. For example, a bank manager may need to determine whether the credit risk of a customer depends on their home ownership status. Credit risk can be measured as good or poor, while home ownership can be measured as renting or buying a home. Since both variables are categorical, the techniques we have learned so far cannot be used to analyse this problem. We therefore have to use the chi-square test of independence.

12.1 Chi-square goodness-of-fit test

LEARNING OBJECTIVE 12.1 Understand the chi-square goodness-of-fit test, use it to test whether a given distribution is similar to a specified distribution (such as the Poisson distribution or normal distribution) and use it to test a hypothesis about a population proportion.

In the binomial distribution, only two possible outcomes can occur on a single trial in an experiment. An extension of the binomial distribution is the **multinomial distribution**, in which there can be more than two possible outcomes in a single trial. Suppose an electronics manufacturer wants to undertake an online survey of customer satisfaction with a particular brand of its products in a particular city. The respondents are to indicate their level of satisfaction by selecting a single number on a scale of 1 (very dissatisfied) to 5 (very satisfied). Suppose that 100 customers are randomly selected to participate in the survey. This is a multinomial experiment consisting of 100 identical trials, where each trial determines the satisfaction of each respondent. In each trial, only one outcome is possible because a respondent is not allowed to select more than one level of satisfaction. The trials are also independent because one respondent's selection does not affect that of any other respondent. The chi-square goodness-of-fit test can be used to determine whether the sample data conform to any kind of expected distribution. As the name suggests, the test determines whether the data 'fit' a given distribution. Suppose, in this example, the electronics company has already conducted a national survey and now wants to know whether customer satisfaction in this city has a pattern similar to the national pattern of satisfaction.

The chi-square goodness-of-fit test compares the *expected*, or theoretical, frequencies of categories from a population distribution with the *observed*, or actual, frequencies from a sample distribution to determine whether there is a significant difference between what is expected and what is observed. In the customer satisfaction example, the sales manager might hypothesise that, because the age profile of the customers in this city is similar to that of the general population, the pattern of customer satisfaction is similar to the national pattern. To either validate or reject this expected distribution, the observed results can be compared with the expected results using the chi-square goodness-of-fit test. This test can also be used to determine whether the customer satisfaction levels are normally distributed or Poisson distributed.



Formula 12.1 is used to compute a chi-square goodness-of-fit test.

**Chi-square
goodness-of-fit test**

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

$$df = k - 1 - m$$

12.1

where:

f_o = the frequency of observed values

f_e = the frequency of expected values

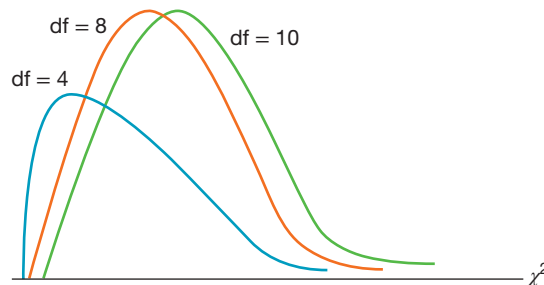
k = the number of categories

m = the number of parameters being estimated
from the sample data

The idea behind the chi-square goodness-of-fit test is to compare how far the actual *observed* frequency is from the *expected* frequency. If the sales manager's hunch that customer satisfaction in the city is similar to national customer satisfaction is correct, we expect the squared difference between f_o and f_e to be zero or very small. Note that you could also compare these by looking at the absolute difference between the two but, because some deviations will be positive and others will be negative, the former might cancel out the latter and the sum of the deviations come to zero. So, as we have learned in calculating the variance and the standard deviation, we square the deviations to ensure that the total deviation is positive. How significant the difference is between f_o and f_e is determined by comparing the test statistic with the critical value at a given level of significance. The degrees of freedom are obtained by first subtracting 1 from the sample size (n), as the case for the t statistic. However, in addition we also subtract m , the number of population parameters of the expected distribution that have been estimated from the data. This is explained in the example below and in demonstration problem 12.1.

Looking at the numerator of the chi-square test statistic we can see that, because we are squaring the deviations between f_o and f_e , the value of χ^2 can never be negative. As with the t distribution, the chi-square distribution belongs to a family of distributions, with each distribution uniquely defined according to its degrees of freedom (df). As can be seen from figure 12.1, the chi-square distribution is considerably skewed to the right (i.e. it has a strong positive skew) at low values of df. However, as df increases (i.e. as n gets larger) the chi-square distribution becomes more and more normal, as with the t distribution. Values for the chi-square distribution are given in table A.8 in the appendix. Because of space limitations, chi-square values are listed only for certain probabilities.

FIGURE 12.1 Family of chi-square distributions



The following example shows how the chi-square goodness-of-fit test can be applied to business situations. As part of a computer manufacturer's commitment to total quality control, it carefully monitors defective components because this provides useful information for quality improvement. Its defective components are placed into three categories according to the type of problem: defective chip, defective soldering point and defective circuit board. Based on past data collected from the assembly line, the quality-control manager knows the manufacturing process is under control if the distribution of defects is as follows: 14.2% defective chips, 60.5% defective soldering points and 25.3% defective circuit boards.

To test whether the process in a given week is under control, the manager randomly selects 50 defective components. The distribution of defects is shown in table 12.1. The manager can now use a chi-square goodness-of-fit test to determine whether the observed frequencies of responses from this survey are the same as the frequencies that would be expected if the quality-control system is under control.

TABLE 12.1 Results of the quality-control survey

Type of defect	Frequency (f_o)
Chip	9
Soldering point	35
Circuit board	6
Total	50

The test is carried out using the following steps.

Step 1: Set up H_0 and H_a

H_0 : The observed distribution is the same as the expected distribution.

H_a : The observed distribution is not the same as the expected distribution.

Step 2: Decide on the type of test

The chi-square goodness-of-fit test is always one-tailed (or one-sided) because a chi-square value of zero (or close to zero) indicates that the observed and expected distributions are similar. Any deviation from a zero difference occurs only in the positive direction because chi-square is determined by a sum of squared values and can never be negative.

Step 3: Decide on the level of significance α and determine the critical value(s) and region(s)

Let $\alpha = 0.05$. Given that there are three categories in this example (defective chip, defective soldering point and defective circuit board), $k = 3$. The value of m is 0 because the test does not require us to use the data to estimate any parameter values when using χ^2 as a goodness-of-fit statistic. Therefore, the degrees of freedom are $k - 1 - m = 3 - 1 - 0 = 2$. For $\alpha = 0.05$ and $df = 2$, the critical chi-square value from table A.8 in the appendix (reading from the second row and going across to the $\alpha = 0.05$ column) is:

$$\chi^2_{0.05, 2} = 5.9915$$

Step 4: Write down the decision rule

If the chi-square value calculated from the survey data (i.e. the observed chi-square) is greater than 5.9915, then we reject the null hypothesis.

Step 5: Select a random sample and do relevant calculations

The sample information (observed frequencies) was given in table 12.1. To compute the chi-square goodness-of-fit test statistic, we first calculate the expected frequencies. The expected proportions are given below as 0.142, 0.605 and 0.253. The observed values from the sample data reported in table 12.1 sum to 50 (i.e. $n = 50$). Under the null hypothesis, the observed distribution is the same as the expected distribution. On that basis, we can calculate the expected frequencies by multiplying the expected proportions by the sample total of the observed frequencies ($n = 50$), as shown in table 12.2.

TABLE 12.2 Expected values for the quality-control problem

Type of defect	Expected proportion	Expected frequency (f_e) (proportion $\times$ sample total)
Chip	0.142	$0.142 \times 50 = 7.10$
Soldering point	0.605	$0.605 \times 50 = 30.25$
Circuit board	0.253	$0.253 \times 50 = 12.65$
Total		50.00

The chi-square goodness-of-fit test statistic is computed in table 12.3. This shows that the observed chi-square value is 4.75. We compare this with the critical value of 5.9915. In figure 12.2, we reject H_0 if the computed test statistic falls in the rejection zone.

TABLE 12.3 Calculation of chi-square for the quality-control problem

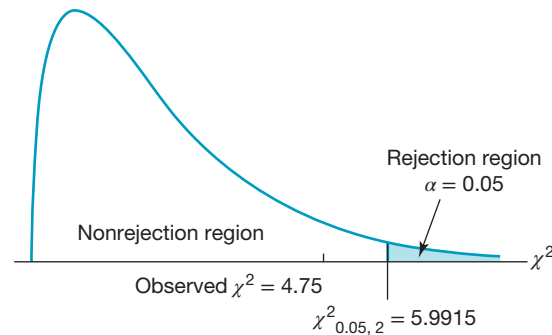
Type of defect	f_o	f_e	
Chip	9	7.10	0.508
Soldering point	35	30.25	0.746
Circuit board	6	12.65	3.496
Total	50	50.00	$\chi^2 = 4.750$

Step 6: Draw a conclusion

Because the observed chi-square of 4.75 is less than the critical value of 5.9915, we fail to reject the null hypothesis that the observed distribution is the same as the expected distribution. It is possible the

assembly line is producing defective components that match the historical ‘in control’ distribution and therefore the observed differences between the two distributions could be due to chance. In this case, we do not have overwhelming evidence that the system is out of control and therefore we accept the possibility that it is still in control. The test result shows that the observed distribution is not significantly different from the expected distribution. On the basis of this evidence, we recommend to the quality-control manager that there is no evidence the process has gone out of control and therefore no corrective measures are necessary.

FIGURE 12.2 Graph of chi-square distribution for quality-control survey



DEMONSTRATION PROBLEM 12.1

Chi-square test to determine significant change

Problem

Roy Morgan Market Research regularly conducts surveys of registered Australian voters to determine their voting intentions. The following table shows the results of an online survey that asked voters which party they would vote for if an election were to be held on that day.

Party	Number
Liberal Party	241
National Party	14
Australian Labor Party	187
Greens	54
Others/independents	46
Total	542

A previous survey indicated the following distribution of voters' preferences: Liberal Party 38%; National Party 5%; Australian Labor Party 38%; Greens 13.5%; Others/independents 5.5%. Determine whether there has been a significant change in voters' preferences using $\alpha = 0.01$.

Solution

Step 1: Set up H_0 and H_a

H_0 : Voters' preferences have not changed; that is, voters' preferences in the recent poll are the same as in the previous poll.

H_a : Voters' preferences have changed; that is, voters' preferences in the recent poll are not the same as in the previous poll.

Step 2: Decide on the type of test

The chi-square goodness-of-fit test is one-tailed.

Step 3: Decide on the level of significance α and determine the critical value(s) and region(s)

Let $\alpha = 0.01$. With four categories in this example, $k = 5$. The value of m is 0 because we did not use the data to estimate any parameters. Therefore, the degrees of freedom are $k - 1 - m = 4 - 1 - 0 = 3$. For $\alpha = 0.01$ and $df = 3$, the critical chi-square value from table A.8 in the appendix is:

$$\chi^2_{0.01,3} = 11.3449$$

Step 4: Write down the decision rule

If the chi-square calculated from the recent survey data (i.e. the observed chi-square) is greater than 11.3449, we reject the null hypothesis.

Step 5: Select a random sample and do relevant calculations

The expected proportions are those from the earlier poll, which were given above as 0.38, 0.05, 0.38, 0.135 and 0.055. The observed values from the sample data reported in the table previously add up to 542. We can therefore calculate the expected frequencies by multiplying the expected proportions by the sample total of the observed frequencies ($n = 542$), as shown in the table below.

Party	Expected proportion	Expected frequency (f_e) (proportion $\times$ sample total)
Liberal Party	0.38	$0.38 \times 542 = 205.96$
National Party	0.05	$0.05 \times 542 = 27.10$
Australian Labor Party	0.38	$0.38 \times 542 = 205.96$
Greens	0.135	$0.135 \times 542 = 73.17$
Others/independents	0.055	$0.055 \times 542 = 29.81$
Total		542.00

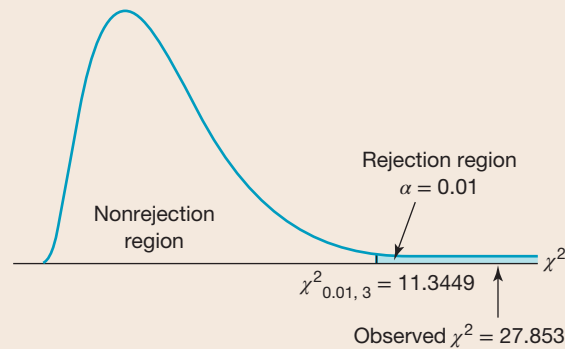
The chi-square test statistic can then be calculated using the table below.

Party	f_o	f_e	
Liberal Party	241	205.96	5.961
National Party	14	27.10	6.332
Australian Labor Party	187	205.96	1.745
Greens	54	73.17	5.022
Others/independents	46	29.81	8.793
			$\chi^2 = 27.853$

From the table above, our observed chi-square is 27.853. We compare this with the critical value of 11.3449. In the graph below, we reject H_0 if the computed test statistic falls in the rejection region.

Step 6: Draw a conclusion

Given that the observed chi-square of 27.853 is greater than the critical value of 11.3449 in the following graph, we reject the null hypothesis and conclude that there has been a significant shift in voter preferences since the previous opinion poll. Specifically, there has been a significant shift from the Australian Labor Party to the Liberal Party and the National Party.



DEMONSTRATION PROBLEM 12.2

Chi-square test to determine Poisson distribution

Problem

The chi-square goodness-of-fit test can also be used to test whether a given distribution has a Poisson distribution. A Poisson distribution is identified by an average number of occurrences of events (λ , lambda) at a given location within a specified time period. This makes it useful for testing hypotheses about events such as waiting times in queues, the number of flaws in a manufactured item and so on. Suppose the traffic manager of a city council believes traffic accidents at a busy intersection are Poisson distributed and sets out to test this hypothesis by gathering information. Confirmation of this distribution will help her to make a more informed recommendation to the council regarding what response should be taken to deal with the problem. The following data represent a distribution of the frequency of traffic collisions at the location during daily intervals within the study period.

Use $\alpha = 0.05$ to test whether the traffic manager's belief is supported by the data.

Number of accidents	Observed frequencies
0	9
1	22
2	28
3	19
≥ 4	6

Solution

Step 1: Set up H_0 and H_a

H_0 : The frequency distribution is Poisson distributed.

H_a : The frequency distribution is not Poisson distributed.

Step 2: Decide on the type of test

The chi-square goodness-of-fit test is one-tailed.

Step 3: Decide on the level of significance α and determine the critical value(s) and region(s)

Let $\alpha = 0.05$. With five categories of accident occurrences, $k = 5$. The value of m is 1 because the mean collision occurrence (λ) was not given and needs to be estimated from the data. Therefore, the degrees of freedom are $k - 1 - m = 5 - 1 - 1 = 3$. For $\alpha = 0.05$ and $df = 3$, the critical chi-square value from table A.8 in the appendix is $\chi^2_{0.05, 3} = 7.8147$.

Step 4: Write down the decision rule

If the chi-square value calculated from the data is greater than 7.8147, we reject the null hypothesis that the distribution is Poisson distributed.

Step 5: Select a random sample and do relevant calculations

We can calculate the expected frequencies in two steps. First, we compute the mean collision occurrence (λ) for the sample. The sample mean is actually a weighted average, which is obtained by multiplying each accident category number by the observed frequency in that category, adding up the product of the number of accidents and the frequency of those accidents, and dividing by the total number of observed frequencies; that is:

$$\lambda = \frac{0(9) + 1(22) + 2(28) + 3(19) + 4(6)}{84} = \frac{159}{84} = 1.89 \approx 1.9$$

Using $\lambda = 1.9$ as our estimate of the population mean, we now refer to the column for $\lambda = 1.9$ in the Poisson probabilities tables in table A.2 in the appendix and read the probabilities of the various categories, where x = the number of collisions. For example, for zero collisions ($x = 0$) the probability is 0.1496; for one collision ($x = 1$) the probability is 0.2842. The remaining probabilities are given in the table below.

In the second step (see column three of the table), we compute the expected frequencies by multiplying the total number of collisions by the expected probabilities (column 2).

Accidents (x)	Expected probability	Expected frequency (f_e) (accidents $\times$ probability)
0	0.1496	$84 \times 0.1496 = 12.57$
1	0.2842	$84 \times 0.2842 = 23.87$
2	0.2700	$84 \times 0.2700 = 22.68$
3	0.1710	$84 \times 0.1710 = 14.36$
≥ 4	0.1252	$84 \times 0.1252 = 10.52$
Total		84.00

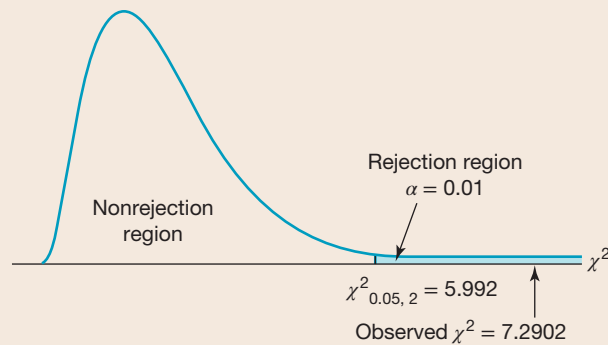
The chi-square test statistic can then be calculated using the following table.

Accidents	Observed frequencies	Expected frequencies	$\frac{(f_o - f_e)^2}{f_e}$
0	7	12.57	1.01
1	18	23.87	0.15
2	25	22.68	1.25
3	17	14.36	1.50
4	12	10.52	1.94
Total	84	84.00	$\chi^2 = 5.85$

From the table above, our observed chi-square is 5.85. We compare this with the critical value of 7.8147. On the following graph, we reject H_0 if the computed test statistic falls in the rejection region.

Step 6: Draw a conclusion

Because the observed chi-square of 5.992 is less than the critical value of 7.2902, we cannot reject the null hypothesis. That is, there is insufficient evidence to disprove the manager's claim that the frequency of collisions at the intersection follows a Poisson distribution.



DEMONSTRATION PROBLEM 12.3

Chi square test to determine normal distribution

Problem

Another application of the chi-square goodness-of-fit test is to determine whether or not a given distribution conforms to a normal distribution. (For continuous data, testing whether or not the data conform to a normal distribution could also be done using the Kolmogorov–Smirnov test.) As an example, consider a business statistics lecturer who is interested in knowing whether the distribution of scores in her class is normal. This information is necessary because she wants to find out what proportion of the students in the course may have special needs. Suppose the lecturer has collected data consisting of a simple random sample of 300 scores in her first-year business statistics course. A previous study determined that the average score in the course was 52.7% with a standard deviation of 15.0%. A frequency distribution of the sample scores is found below.



Score (%)	Frequency
Under 30	10
30–under 40	35
40–under 50	97
50–under 60	78
60–under 70	38
70–under 80	30
80–under 90	8
90–under 100	4
Total	300

Based on the sample data, determine at the 0.01 level of significance whether the sample could have been drawn from a population in which the scores are normally distributed.

Solution

Step 1: Set up H_0 and H_a

H_0 : The sample was drawn from a population of scores that are normally distributed.

H_a : The sample was drawn from a population of scores that are not normally distributed.

Step 2: Decide on the type of test

The chi-square goodness-of-fit test is appropriate for testing this hypothesis.

Step 3: Decide on the level of significance α and determine the critical value(s) and region(s)

Let $\alpha = 0.01$. With eight categories of scores, $k = 8$. The value of m is 0 because there was no need to estimate any parameters; the mean and standard deviation are given. Therefore, the degrees of freedom are $k - 0 - m = 8 - 0 - 1 = 7$. For $\alpha = 0.01$ and $df = 7$, the critical chi-square value from table A.8 in the appendix is $\chi^2_{0.01, 7} = 18.4753$.

Step 4: Write down the decision rule

Reject the null hypothesis if the chi-square value calculated from the data is greater than 18.4753.

Step 5: Select a random sample and do relevant calculations

Under the null hypothesis, the distribution has a normal distribution. We can therefore calculate the expected frequencies as follows. First, we use the given mean and standard deviation to calculate the z-scores at the respective upper class limits (x). We then use table A.4 in the appendix to find the areas under the standard normal curve to the left of each x . Next, we find the area or probability (p) for each class interval. Finally, we multiply each of these probabilities by the total number of observations to obtain the expected frequencies. If the expected frequency in any is less than five, we combine that class with an adjacent one that has more than five.

As an example, for the 'under 30' class the z-score is calculated as follows.

$$z = \frac{30 - 52.7}{15} = -1.51$$

The area to the left of $z = -1.51$ is 0.065. The area of this class interval is also 0.065. The expected frequency in this class is given by $300p = 300(0.065) = 19.5$.

For the '30-under 40' class the z-score is:

$$z = \frac{40 - 52.7}{15} = -0.85$$

The area to the left of $z = -0.85$ is 0.198. The area of this class interval is given by $0.198 - 0.065 = 0.133$. The expected frequency is given by $300p = 300(0.133) = 39.9$. The remaining expected frequencies are shown in the table below.

Upper class limit (x)	$z = \frac{x - \mu}{\sigma} = \frac{x - 52.7}{15}$	Area under standard normal curve left of x	Area of class interval (p)	Expected frequency, $f_e = 300p$
30	-1.51	0.065	0.065	19.5
40	-0.85	0.198	0.133	39.9
50	-0.18	0.429	0.231	69.3
60	0.49	0.688	0.259	77.7
70	1.15	0.875	0.187	56.1
80	1.82	0.966	0.091	27.3
90	2.49	0.994	0.028	8.4
100	3.15	1.000	0.006	1.8
Total			1.000	300.0

The chi-square test statistic can then be calculated using the following table.

Upper class limit (x)	Observed frequencies (f_o)	Expected frequencies (f_e)	$\frac{(f_o - f_e)^2}{f_e}$
30	10	19.5	4.63
40	35	39.9	0.60
50	97	69.3	11.07
60	78	77.7	0.00
70	38	56.1	5.84
80	30	27.3	0.27
90	8	8.4	0.02
100	4	1.8	2.69
Total	300	300.0	$\chi^2 = 25.12$

From the above table, our observed chi-square value is 25.12. This is greater than the critical value of 18.4753.

Step 6: Draw a conclusion

Because the observed chi-square value of 25.12 is greater than the critical value of 18.4753, we reject the null hypothesis; that is, the sample was not drawn from a population with a normal distribution.

DEMONSTRATION PROBLEM 12.4

Chi-square test for population proportions

A final application of the chi-square goodness-of-fit test is to test hypotheses about a population proportion. This can be used as an alternative technique to the z test. Using the z test of a population proportion, when the sample size is large enough ($np \geq 5$ and $nq \geq 5$) sample proportions are normally distributed and the following formula can be used to test hypotheses about p :

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

The chi-square goodness-of-fit test can also be used to conduct tests about p ; in this simple case, it can be viewed as a special case of the chi-square goodness-of-fit test where the number of classifications equals two (binomial distribution). The observed chi-square is computed in the same way as in any other chi-square goodness-of-fit test but, because the test contains only two classifications (success or failure), $k = 2$ and the degrees of freedom are $k - 1 = 2 - 1 = 1$.

As an example, we solve the following problem from section 9.5 by using the chi-square methodology. This example tests the hypothesis that exactly 8% of a manufacturer's products are defective. The following hypotheses are being tested.

$$H_0: p = 0.08$$

$$H_a: p \neq 0.08$$

The value of alpha was given to be 0.10. To test these hypotheses, a business researcher randomly selected a sample of 200 items and determined that 33 of the items had at least one flaw.

In solving this problem using the chi-square goodness-of-fit test, we view it as a two-category expected distribution in which we expect 0.08 defective and 0.92 nondefective items. The observed categories are

33 defective and $200 - 33 = 167$ nondefective. Using the total number of observed items (200), we can determine the expected distributions as $0.08(200) = 16$ and $0.92(200) = 184$. Shown here are the observed and expected frequencies.

	f_o	f_e
Defective	33	16
Nondefective	167	184

Alpha is 0.10, the test is two-tailed and there is one degree of freedom. The critical table chi-square value is:

$$\chi^2_{0.10,1} = 2.7055$$

An observed chi-square value greater than this value must be obtained to reject the null hypothesis. The chi-square for this problem is calculated as follows.

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} = \frac{(33 - 16)^2}{16} + \frac{(167 - 184)^2}{184} = 18.06 + 1.57 = 19.63$$

Note that this observed value of chi-square, 19.63, is greater than the critical table value, 2.7055. The decision is therefore to reject the null hypothesis; according to this analysis, the manufacturer does not produce 8% defective products. Observing the actual sample result in which 0.165 of the sample was defective indicates that the proportion of the population that is defective might be greater than 8%.

Note also that the above results are approximately the same as those calculated using the z test. This can be explained by the interrelationship between the standardised normal distribution and a chi-square distribution with one degree of freedom. For example, using the z test the observed z value is 4.43 and here our observed χ^2 value is 19.63. Note that $\chi^2 \approx z^2$; that is, $19.63 \approx 4.43^2$. This relationship also applies to the critical values of the two test statistics at the 0.10 level of significance. For example, the critical value of χ^2 with one degree of freedom at $\alpha = 0.10$ is 2.7055, which is approximately equal to the square of the critical z value of 1.645. Furthermore, looking at the $df = 1$ row in table A.8 in the appendix, it can be seen that observed chi-square statistic of 19.63 has a p -value of less than 0.005, which is close to zero and roughly equal to the p -value of zero of the observed z statistic. This shows that, when doing a two-tailed test for the equality of two proportions, the z and the χ^2 tests are equivalent. However, if we are specifically interested in a one-tailed test, we must use the z test with the entire rejection region put into one tail of the standardised normal distribution.

PRACTICE PROBLEMS

Using the chi-square test

Practising the calculations

12.1 Use a chi-square goodness-of-fit test to determine whether the observed frequencies have the same distribution as the expected frequencies ($\alpha = 0.05$).

Category	f_o	f_e
1	36	46
2	96	66
3	78	83
4	25	30
5	17	11

- 12.2** Are the following data Poisson distributed? Use $\alpha = 0.05$ and the chi-square goodness-of-fit test to answer this question. What is your estimated lambda?

Number of arrivals	f_o
0	28
1	17
2	11
3	5

- 12.3** Use the chi-square goodness-of-fit test to determine if the following observed data are normally distributed. Let $\alpha = 0.05$. What are the estimated mean and standard deviation?

Category	Observed
1 to less than 10	6
11 to less than 20	14
21 to less than 30	29
31 to less than 40	38
41 to less than 50	25
51 to less than 60	10
61 to less than 70	7

- 12.4** In one survey, female entrepreneurs were asked to select their personal definition of success from several categories. Thirty-nine per cent responded that happiness was their definition of success, 12% said sales/profit, 18% said helping others and 31% said achievements/challenge. Suppose you wanted to determine whether male entrepreneurs felt the same way and took a random sample of men, resulting in the following data. Use the chi-square goodness-of-fit test to determine whether the observed frequency distribution of data for men is the same as the distribution for women. Let $\alpha = 0.05$.

Definition	f_o
Happiness	42
Sales/profit	95
Helping others	27
Achievements/challenge	63

Testing your understanding

- 12.5** The general manager of a cosmetics marketing company believes the annual incomes of its clients are normally distributed. The following data represent the distribution of incomes for a random sample of the company's clients. Given a sample mean of \$67 000 and a standard deviation of \$24 000, use the chi-square goodness-of-fit test to determine whether this distribution is significantly different from the normal distribution. Assume that $\alpha = 0.05$. How could the result help the company to sell more cosmetics?

Annual income of client (\$)	Frequency
0–20 000	2
20 001–50 000	20
50 001–80 000	51
80 001–110 000	22
110 001–140 000	5

12.6 An emergency call service centre keeps records of emergency telephone calls. A study of 150 five-minute time intervals resulted in the following distribution of number of calls. For example, during 18 of the five-minute intervals, no calls were received. Use the chi-square goodness-of-fit test and $\alpha = 0.01$ to determine whether this distribution is Poisson. Explain your test result.

Number of calls (per 5-minute interval)	Frequency
0	18
1	28
2	47
3	21
4	16
5	11
6 or more	9

12.7 According to a recent survey, 66% of all computer companies are going to spend more on marketing this year than in previous years. Only 33% of other IT companies and 28% of non-IT companies are going to spend more than in previous years. Suppose a researcher wanted to conduct a survey of her own to test the claim that 28% of all non-IT companies will spend more on marketing next year than this year. She randomly selects 270 companies and determines that 62 of the companies do plan to spend more on marketing next year. Use $\alpha = 0.05$, the chi-square goodness-of-fit test and the sample data to determine whether the 28% figure holds for all non-IT companies.

12.2 Contingency analysis: chi-square test of independence

LEARNING OBJECTIVE 12.2 Understand the chi-square test of independence and use the chi-square test of independence to determine whether two variables are independent.

Hypothesis tests involving one and two population proportions are useful in many cases, but we will also encounter business situations where there is a need to test hypotheses involving many proportions. For example, the manager of a superannuation fund that offers five different packages may wish to determine whether the proportion of customers selecting each package is related to the four sales regions in which the customers reside. Another example is that of a campaign manager of a political party who wants to know whether party affiliation is related to educational level. In each of these cases, the proportions relate to characteristic categories of the variable of interest.

The goodness-of-fit test cannot be used to analyse these situations that involve two or more categorical variables. Such cases require a new statistical technique called the chi-square test of independence or

contingency analysis. Contingency analysis is based on a **contingency table** and tests whether two events are independent. Here we use the chi-square test of independence to determine whether two variables are related.

Suppose a business researcher is interested in determining whether investors' state or territory of residence is independent of the type of financial investment they have. On the survey questionnaire, the following two questions might be used to measure residence and type of financial investment:

In which state/territory do you reside?

1. New South Wales
2. Victoria
3. South Australia
4. Queensland
5. Northern Territory
6. Western Australia
7. Australian Capital Territory
8. Tasmania

Which type of financial investment are you most likely to make today?

1. Stocks
2. Government securities
3. Other

The business researcher would tally the frequencies of responses to these two questions in a two-way contingency table. Table 12.4 shows a contingency table for these two variables.

TABLE 12.4 Contingency table for the investment example

State/territory	Type of financial investment		
	Stocks	Government securities	Other
New South Wales			o_{13}
Victoria			
South Australia			
Queensland			
Northern Territory			
Western Australia			
Australian Capital Territory			
Tasmania	o_{81}		

The following notation is used to describe entries in the table. Subscript i denotes the row and j the column. Thus, o_{13} is the observed frequency for the cell in the first row and third column of data, and o_{81} is the observed frequency for the cell in the eighth row and first column of data. The expected frequencies are denoted in a similar manner. The expected frequencies are calculated using:

$$e_{ij} = \frac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{n}$$

where:

e_{ij} = the expected frequency in row i , column j

n = the total of all the frequencies (i.e. sum of the rows and columns).

Using these expected frequency values and the observed frequency values, we can compute a chi-square test of independence to determine whether the variables are independent using formula 12.2.

Chi-square test of independence

$$\chi^2 = \sum_r \sum_c \frac{(f_o - f_e)^2}{f_e}$$

$$df = (r - 1)(c - 1)$$

where:

r = the number of rows

c = the number of columns

12.2

Note that the chi-square test of independence is one-tailed because we are again comparing f_o and f_e . If H_a is true, we would expect the absolute difference between the two to be significantly greater than zero. Also note that formula 12.2 is similar to formula 12.1, with the exception that the values are summed across both rows and columns, and the degrees of freedom are different.

To illustrate this test, consider an insurance-claims analyst who examines 100 motor vehicle insurance claims to determine whether the gender of a claimant is independent of the claimant's age. She groups the ages of the claimants into the following three categories: less than 30 years, 30–50 years and more than 50 years. The resulting 3×2 contingency table is shown in table 12.5. Using $\alpha = 0.05$, she can use the chi-square test of independence to determine whether gender is independent of age.

TABLE 12.5 Contingency table for the insurance-claims example

Age	Gender		Total
	Male	Female	
<30 years	32	25	57
30–50 years	18	5	23
>50 years	17	3	20
Total	67	33	100

Step 1: Set up H_0 and H_a

H_0 : Gender is independent of age.

H_a : Gender is not independent of age.

Step 2: Decide on the type of test

The chi-square test of independence is appropriate.

Step 3: Decide on the level of significance α and determine the critical value(s) and region(s)

Let $\alpha = 0.05$. Here, there are three rows ($r = 3$) and two columns ($c = 2$). The degrees of freedom are $(3 - 1)(2 - 1) = 2$. The critical value of chi-square for $\alpha = 0.05$ from table A.8 in the appendix is 5.9915.

Step 4: Write down the decision rule

If the chi-square value calculated from the data is greater than 5.9915, we reject the null hypothesis and conclude that gender is not independent of age.

Step 5: Select a random sample and do relevant calculations

The process of computing the test statistic begins by calculating the expected frequencies using the formula for e_{ij} given previously. This formula multiplies the frequency of one category by the frequency of the other and divides the product by the total sample size (n). Note that the numerator is similar to the multiplication rule for independent events. This rule says that, when two events are independent, the

probability that they will *both* occur is equal to the product of their probabilities. Therefore, this equation for defining expected frequency expresses independence, in effect, by multiplying these probabilities.

$$e_{11} = \frac{(\text{row 1 total})(\text{column 1 total})}{\text{grand total}} = \frac{57(67)}{100} = 38.19$$

$$e_{12} = \frac{(\text{row 1 total})(\text{column 2 total})}{\text{grand total}} = \frac{57(33)}{100} = 18.81$$

$$e_{21} = \frac{(\text{row 2 total})(\text{column 1 total})}{\text{grand total}} = \frac{23(67)}{100} = 15.41$$

$$e_{22} = \frac{(\text{row 2 total})(\text{column 2 total})}{\text{grand total}} = \frac{23(33)}{100} = 7.59$$

$$e_{31} = \frac{(\text{row 3 total})(\text{column 1 total})}{\text{grand total}} = \frac{20(67)}{100} = 13.40$$

$$e_{33} = \frac{(\text{row 3 total})(\text{column 3 total})}{\text{grand total}} = \frac{20(33)}{100} = 6.60$$

We then list the expected frequencies in the cells of the contingency table (table 12.6) along with the observed frequencies. The expected frequencies are shown in parentheses beneath the observed frequencies.

TABLE 12.6 Contingency table of observed and expected frequencies (in brackets) for the insurance-claims example

Age	Gender		Total
	Male	Female	
<30 years	32 (38.19)	25 (18.81)	57
30–50 years	18 (15.41)	5 (7.59)	23
>50 years	17 (13.40)	3 (6.60)	20
Total	67	33	100

Next, we compute the chi-square value by summing $\frac{(f_o - f_e)^2}{f_e}$ for all cells.

$$\begin{aligned}\chi^2 &= \frac{(32 - 38.19)^2}{38.19} + \frac{(25 - 18.81)^2}{18.81} + \frac{(18 - 15.41)^2}{15.41} + \frac{(5 - 7.59)^2}{7.59} \\ &\quad + \frac{(17 - 13.4)^2}{13.4} + \frac{(3 - 6.6)^2}{6.6} \\ &= 1.0033 + 2.0370 + 0.4353 + 0.8838 + 0.9672 + 1.9636 \\ &= 7.2902\end{aligned}$$

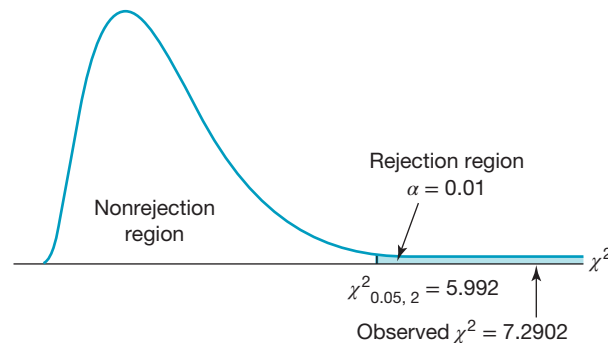
From the preceding calculation, the observed value of chi-square is 7.2902. We then compare this with the critical value of 5.9915. In the following graph, we reject H_0 if the computed test statistic falls in the rejection region.

Step 6: Draw a conclusion

Since the observed chi-square of 7.2902 is greater than the critical value of 5.9915, we reject the null hypothesis and conclude that the gender of an insurance claimant is not independent of the claimant's

age. Looking at the proportions in the cells, it can be seen that, in all age groups, women make fewer claims than men.

FIGURE 12.3 Graph of chi-square distribution for insurance-claims example



DEMONSTRATION PROBLEM 12.5

Chi-square test of independence

Problem

A popular supermarket chain routinely asks its customers whether they prefer plastic bags or their own bags with their purchases. A recent survey of customers recorded information on their bag preferences as well as their educational background. Using the information from the table below, can we conclude at the 0.01 level of significance that there is a relationship between educational background and type of bag preferred?

Bag preference	Educational background		
	High school	Tertiary	Postgraduate
Own bag	20	40	80
Plastic bag	90	20	15
No preference	20	12	12

Solution

Step 1: Set up H_0 and H_a

H_0 : Bag preference is independent of educational background.

H_a : Bag preference is not independent of educational background.

Step 2: Decide on the type of test

Because we want to test the relationship between two variables (bag preference and educational background) and the data are frequencies, the chi-square test of independence is appropriate.

Step 3: Decide on the level of significance α and determine the critical value(s) and region(s)

The degrees of freedom are $(3 - 1)(3 - 1) = 4$ and the critical value is $\chi^2_{0.01, 4} = 13.2767$.

Step 4: Write down the decision rule

If the chi-square value calculated from the data is greater than 13.2767, we reject the null hypothesis and conclude that bag preference is not independent of educational background.

Step 5: Select a random sample and do relevant calculations

The expected frequencies are the products of the row and column totals divided by the grand total. The contingency table, with observed and expected frequencies (in brackets), follows.

Bag preference	Educational background			Total
	High school	Tertiary	Postgraduate	
Own bag	20 (58.90)	40 (32.62)	80 (48.48)	140
Plastic bag	90 (52.59)	20 (29.13)	15 (43.28)	125
No preference	20 (18.51)	12 (10.25)	12 (15.24)	44
Total	130	72	107	309

The chi-square value is computed as follows.

$$\begin{aligned}
 \chi^2 &= \frac{(20 - 58.90)^2}{58.90} + \frac{(40 - 32.62)^2}{32.62} + \frac{(80 - 48.48)^2}{48.48} + \frac{(90 - 52.59)^2}{52.59} \\
 &\quad + \frac{(20 - 29.13)^2}{29.13} + \frac{(15 - 43.28)^2}{43.28} + \frac{(20 - 18.51)^2}{18.51} + \frac{(12 - 10.25)^2}{10.25} + \frac{(12 - 15.24)^2}{15.24} \\
 &= 25.691 + 20.493 + 26.612 + 2.862 + 18.479 + 18.479 + 0.123 + 0.298 + 0.689 \\
 &= 95.247
 \end{aligned}$$

Step 6: Draw a conclusion

Because the observed chi-square of 95.247 is greater than the critical value of 13.2767, we reject the null hypothesis. Thus, there is sufficient evidence to conclude that there is a relationship between educational background and type of bag preferred. The relatively high value of the calculated chi-square statistic in relation to the critical value of chi-square suggests that, at $\alpha = 0.01$, the differences between the observed and expected frequencies are large enough to support the view that bag preference is related to customers' educational background.

PRACTICE PROBLEMS**Chi-square test of independence****Practising the calculations**

12.8 Use the following contingency table to determine whether brand preference for breakfast cereal is independent of gender. Let $\alpha = 0.01$.

Gender	Brand preference			Total
	A	B	C	
Male	18	25	17	60
Female	32	5	3	40
Total	50	30	20	100

- 12.9** Use the following contingency table and the chi-square test of independence to determine whether socio-economic status is independent of the number of children in a family. Let $\alpha = 0.05$.

Number of children	Socio-economic status		
	Low	Medium	High
0	7	18	6
1	9	38	23
2 or 3	34	97	58
More than 3	47	31	30

- 12.10** A motor insurance company wants to know whether the frequency of insurance claims by its customers is related to their age. Test the hypothesis at $\alpha = 0.1$ using the information below.

Action	Age of customer			
	Under 25	25–35	36–55	Over 55
Filed claim	85	92	100	65
Did not file claim	92	121	600	140

Testing your understanding

- 12.11** A computer manufacturer is considering sourcing supplies of an electrical component from one of four different manufacturers. The purchasing manager asks for 95 samples from each of the manufacturers. The numbers of acceptable and unacceptable components from each manufacturer are shown in the following table. Use a chi-square test to determine whether the quality of the component depends on the manufacturer. Let $\alpha = 0.05$. How does this help the manager to make decisions?

Quality	Manufacturer				Total
	A	B	C	D	
Unacceptable	12	8	5	11	36
Acceptable	83	87	90	84	344
Total	95	95	95	95	380

- 12.12** An accounting firm studied the time its accountants took to complete each client's tax return and related this to the amount of tax paid by the client. The firm wants to know whether a relationship exists between these two variables, as it is considering a new pricing policy that would depend on the time taken to complete a tax return. The following data were obtained from a survey. Perform an appropriate test using $\alpha = 0.05$. Would the proposed policy be justified based on your results? How could this information be used to improve the processing of tax returns?

No. of work hours	Amount of tax paid		
	\$0–3000	\$3001–10 000	Over \$10 000
0–2	52	55	27
3–4	47	55	31
Over 4	35	37	35

12.13 An investor wants to know whether the distribution of weekly rental values in a certain location is normal. The results of her test are as shown. Discuss the test used, the hypotheses, the results and the business implications. Use $\alpha = 0.05$.

Rental category	Observed frequency (f_o)	Expected frequency (f_e)	($f_o - f_e$)	($f_o - f_e$) ² / f_e
Under \$400	6	4.68	1.32	0.37
\$400–600	14	21.40	–7.40	2.56
\$601–800	45	39.41	5.59	0.79
\$801–1000	30	30.18	–0.18	0.001
Above \$1000	10	9.20		0.07
Total	105	105	0.80	

SUMMARY

12.1 Understand the chi-square goodness-of-fit test, use it to test whether a given distribution is similar to a specified distribution (such as the Poisson distribution or normal distribution) and use it to test a hypothesis about a population proportion.

The chi-square goodness-of-fit test is used to determine whether a given set of data or a situation follows a particular distribution (e.g. normal or Poisson). We use a theoretical or expected distribution of measurements for several categories of a variable with the actual or observed distribution of measurements. If only two categories are used, the test offers the equivalent of a z test for a single proportion. The chi-square goodness-of-fit test for two categories is also equivalent to a z test of the equality of two population proportions against a two-sided alternative.

12.2 Understand the chi-square test of independence and use the chi-square test of independence to determine whether two variables are independent.

The chi-square test of independence is used to investigate whether there is dependence (a relationship) or independence (no relationship) between two categorical variables. The data used to conduct a chi-square test of independence are arranged in a two-dimensional table called a contingency table. For this reason, the test is sometimes referred to as contingency analysis. A chi-square test of independence is computed in a manner similar to that used with the chi-square goodness-of-fit test. Expected values are computed for each cell of the contingency table and then compared with the observed values using the chi-square statistic. Both the chi-square test of independence and the chi-square goodness-of-fit test require that expected values be greater than or equal to five in order to be confident that the results are sound.

KEY TERMS

chi-square goodness-of-fit test A statistical test used to determine whether a sample could have come from a given type of population distribution.

chi-square test of independence A statistical test used to analyse whether there is a relationship (or dependence) between two categorical variables.

contingency analysis Another name for the chi-square test of independence.

contingency table A table that presents the frequencies with which two events occur and co-occur.

multinomial distribution A distribution in which there are more than two possible outcomes in a single trial.

KEY EQUATIONS

Equation	Description	Formula
12.1	Chi-square goodness-of-fit test	$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$ $df = k - 1 - m$
12.2	Chi-square test of independence	$\chi^2 = \sum_r \sum_c \frac{(f_o - f_e)^2}{f_e}$ $df = (r - 1)(c - 1)$

REVIEW PROBLEMS

PRACTISING THE CALCULATIONS

- 12.1** Use a chi-square goodness-of-fit test to determine whether the following observed frequencies represent a uniform distribution. Let $\alpha = 0.01$.

Category	f_o
1	20
2	17
3	13
4	14
5	18
6	21
7	19
8	18

- 12.2** Use the chi-square contingency analysis to determine whether variable 1 is independent of variable 2. Use a 5% level of significance.

Variable 2			
Variable 1	12	23	21
	8	17	20
	7	11	18

- 12.3** Four investment companies were asked in a survey to rate the effect on their asset values of a 1 percentage point fall in stockmarket prices. The effect on asset values was rated using a 3-point scale, with 1 for no effect, 2 for a slight effect and 3 for a moderate effect. Actual (observed) counts of the responses are shown in the following table. Is the effect on asset values independent of the company? Let $\alpha = 0.01$.

Company	Effect on asset values			Total
	None	Slight	Moderate	
A	60	27	7	94
B	71	22	14	107
C	57	38	15	110
D	54	37	18	109
Total	242	124	54	420

- 12.4** The human resources (HR) manager of a large IT company has collected data for days of the week on which employees were absent from work. The manager randomly selects 150 of the absences, which are shown in the table below. Use $\alpha = 0.05$ to determine whether these data indicate that absences on the various days of the week are equally likely.

Day of week	Observed frequency
Monday	42
Tuesday	18
Wednesday	24
Thursday	27
Friday	39
Total	150

- 12.5** A researcher interviewed 2067 people and asked whether they were the primary decision-makers in the household when buying a new car last year. Two hundred and seven were men who bought a new car last year. Sixty-five were women who bought a new car last year. Eight hundred and eleven of the responses were from men who did not buy a car last year. Nine hundred and eighty-four were from women who did not buy a car last year. Use these data to determine whether gender is independent of being a major decision-maker in purchasing a new car last year. Let $\alpha = 0.05$.



TESTING YOUR UNDERSTANDING

- 12.6** Are random arrivals at a shoe store at the local shopping mall Poisson distributed? A mall employee researches this question by gathering data for arrivals during ten-minute intervals on a weekday between 6.30 pm and 8.00 pm. The data obtained follow. Use $\alpha = 0.05$ to determine whether the observed data seem to be from a Poisson distribution. How could this result help decision-making?

Arrivals per ten-minute interval	Observed frequency
0	26
1	40
2	57
3	32
4	17
5	12
6	8

- 12.7** As part of an infrastructure planning exercise, a city council analyst summarised annual household income and the number of motor vehicles owned for a random sample of households. The results are as follows. Using the chi-square test, determine whether the number of vehicles owned depends on the household's income. Use $\alpha = 0.05$. Explain your test result.

Number of vehicles	Household income			
	Under \$20 000	\$20 000–\$49 999	\$50 000–\$99 999	\$100 000 and above
0	20	12	10	8
1–2	9	18	30	80
3 or more	7	13	15	40

- 12.8** Are the types of professional jobs in the computing industry independent of the number of years a person has worked in the industry? Suppose 246 workers are interviewed. Use the results below to determine whether the type of professional job in the computer industry is independent of the number of years worked in the industry. Let $\alpha = 0.01$. What are the implications, if any, of your results for job retention in this industry?

Years	Professional position			
	Manager	Programmer	Operator	Systems analyst
0–3	6	37	11	13
4–8	28	16	23	24
More than 8	47	10	12	19

- 12.9** In the field of finance, the random-walk hypothesis says that stockmarket prices follow what is described as a ‘random walk’. This means that the price change today is expected to be independent of the price change yesterday. The table below shows price changes for a particular stock collected over 100 days. The price changes are categorised into three groups: up, no change and down. Test the hypothesis that the daily price changes for this stock are independent. Use $\alpha = 0.05$.

Price change today	Price change previous day		
	Up	No change	Down
Up	10	15	12
No change	5	10	5
Down	15	12	8

- 12.10** The following is a 3×2 contingency table for annual farm profit and the age of the farmer for a randomly selected sample of farmers in Fiji. Use an appropriate test to determine whether annual farm profit is related to a farmer’s age. Comment on the results of your test.

Age category	Annual farm profit		Total
	<\$2000	\$2000 or more	
Less than 40 years	29	18	47
40–49 years	51	38	89
50 years or more	82	80	162
Total	162	136	298

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